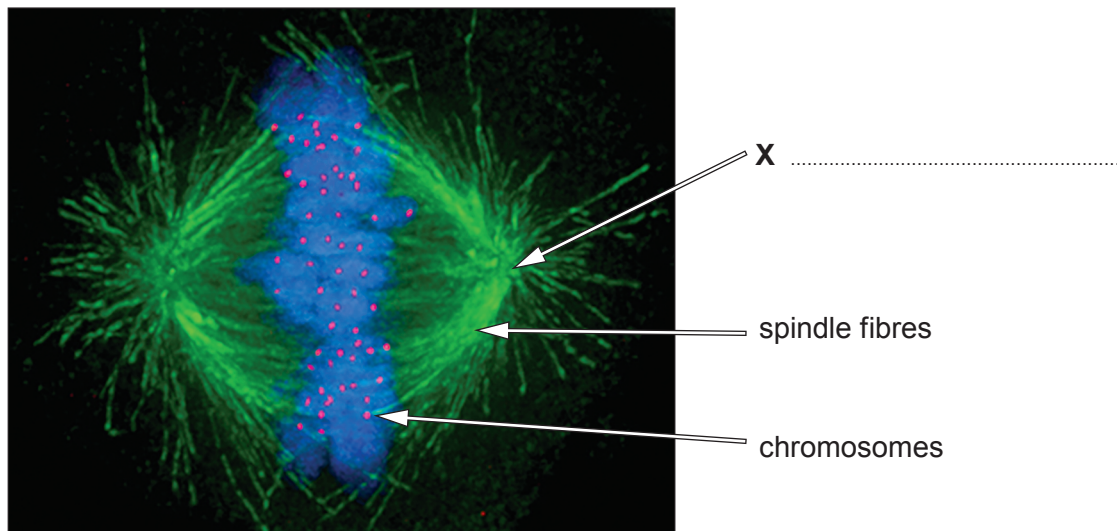


6. **Image 6.1** shows a eukaryotic cell undergoing mitosis. Different structures have been stained with different fluorescent dyes.

Image 6.1



- (a) (i) On **Image 6.1**, name the origin of the spindle fibres, labelled **X**. [1]

- (ii) I. Name the stage of mitosis shown in **Image 6.1**. Explain your reasoning. [2]

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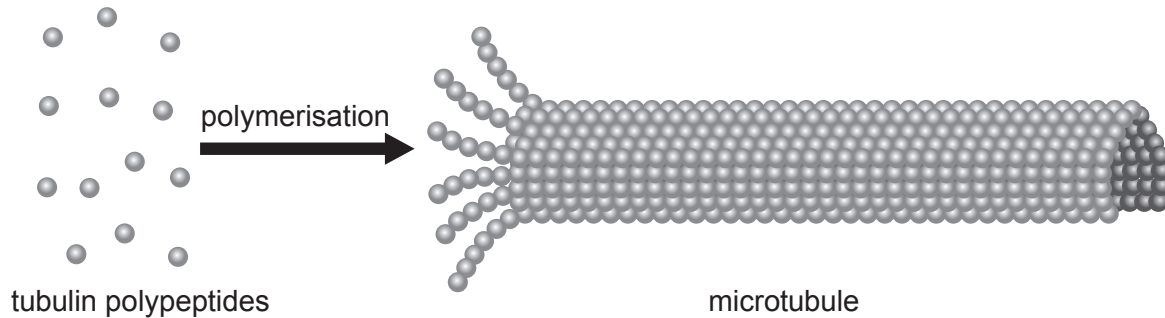
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- II. In the space below **draw a labelled diagram** of **one chromosome** as it would appear at this stage of mitosis. Your diagram should have **two** labels. [2]



- (b) The mitotic spindle is formed at the start of mitosis, it is made up of many small polypeptides called tubulin which are found in the cytoplasm of eukaryotic cells. Before cell division, the tubulin polypeptides must be assembled and joined together to form microtubules. This is shown in **Image 6.2**.

Image 6.2



- (i) Use the information provided to state the highest level of protein structure shown by the microtubule in **Image 6.2**. Explain your answer. [2]

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- (ii) Colchicine is a chemical which prevents the polymerisation of tubulin to form microtubules. Suggest how this would affect the growth of organisms. Explain your answer. [3]

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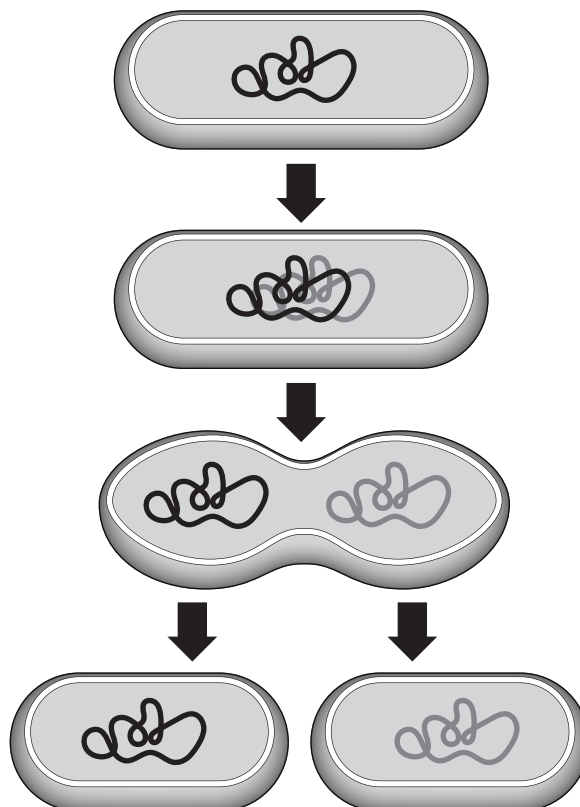
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- (c) When bacteria reproduce, they undergo a process called binary fission. This is shown in **Image 6.3**.

Image 6.3



Use **Image 6.3** and your knowledge of eukaryotic cell division to give **two** reasons why cell division in bacteria is not referred to as mitosis. [2]

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